Exercises

1	2	3	4	5	6	7	8	9	10

Surname, First name

Atay, Kozey

5EZB0 Math 2
Interim test 5EZB0 Q4
21 May, 2026, 15:45-17:15

2	2	4	8	6	7	0

a ~~b~~ c d e f → b
a ~~b~~ c d e f → a
~~a~~ ~~b~~ c d e f → b
~~a~~ ~~b~~ c d e f → d

Answer multiple-choice questions as shown in the example.

Particular Ans on paper exam instructions

- Write in a black or blue pen.
- Hand in all pages. **Do not remove the staple.** If you remove it anyhow, check that you hand in all pages.

Dear student,

This is the midterm test for Math 2 (5EZB0). Write down your name and your student ID at the appropriate places above. Make sure that you enter your student ID by fully coloring the appropriate boxes. On the examination attendance card, you fill in the PDF number. You can find the correct number on the top of the first page of your exam (e.g. 1234.pdf).

Please read the following information carefully:

Date exam: 21/05/2026
Start time: 15:45
End time: 17:15

Number of questions: 10
Maximum number of points/distribution of points over questions: 20 (10 questions, each question 2 points)
Method of determining the final grade: Number of point, divided by 2, rounded to one digit behind the decimal point.
Answering style: multiple choice:

Permitted examination aids

- Scrap paper (fully blank)
- Calculator(**basic**)
- Formula sheet (contained in exam)

Important:

- You are only permitted to visit the toilets under supervision
- Examination scripts (fully completed examination paper, stating name, student number, etc.) must always be handed in
- The house rules must be observed during the examination
- The instructions of subject experts and invigilators must be followed
- Keep your work place as clean as possible: put pencil case and breadbox away, limit snacks and drinks
- You are not permitted to share examination aids or lend them to each other
- Do not communicate with any other person by any means

During written examinations, the following actions will in any case be deemed to constitute fraud or attempted fraud:

- using another person's proof of identity/campus card (student identity card)
- having a mobile telephone or any other type of media-carrying device on your desk or in your clothes
- using, or attempting to use, unauthorized resources and aids, such as the internet, a mobile telephone, smartwatch, smart glasses etc.
- having any paper at hand other than that provided by TU/e, unless stated otherwise
- copying (in any form)
- visiting the toilet (or going outside) without permission or supervision

First-year bachelor students: The final grade for this exam will be announced no later than fifteen working days after the date of this exam, unless this exam takes place in Q4 or the interim period. For Q4 final exams, grades will be announced within five working days after the end of the Q4 final test period. For interim period final exams, grades will be announced no later than five working days before September 1.
All other students: Generally, the final grade for this exam will be announced no later than fifteen working days after the date of this examination. Specifically for bachelor exams administered in the interim period, exam grades will be announced no later than five working days before September 1.

You can start the exam now, good luck!

Formulas

Primitieven / Antiderivatives (zonder/without "$+C$")

$f(x)$	$\int f(x)\,\mathrm{d}x$	$f(x)$	$\int f(x)\,\mathrm{d}x$
x^n, $n \neq -1$	$\frac{x^{n+1}}{n+1}$	$\cos(x)$	$\sin(x)$
$\frac{1}{x}$	$\ln(\lvert x \rvert)$	$\frac{1}{\cos^2(x)}$	$\tan(x)$
e^x	e^x	$\frac{1}{\sqrt{1-x^2}}$	$\arcsin(x)$
a^x, $a > 0$, $a \neq 1$	$\frac{a^x}{\ln(a)}$	$-\frac{1}{\sqrt{1-x^2}}$	$\arccos(x)$
$\sin(x)$	$-\cos(x)$	$\frac{1}{1+x^2}$	$\arctan(x)$

Taylorpolynomials around $x = 0$

Function	Taylorpolynomial plus $\mathcal{O}$ – term
e^x	$1 + x + \frac{1}{2}x^2 + \cdots + \frac{1}{n!}x^n + \mathcal{O}(x^{n+1})$
$\cos(x)$	$1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \cdots + \frac{(-1)^n}{(2n)!}x^{2n} + \mathcal{O}(x^{2n+2})$
$\sin(x)$	$x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots + \frac{(-1)^n x^{2n+1}}{(2n+1)!} + \mathcal{O}(x^{2n+3})$
$\frac{1}{1-x}$	$1 + x + x^2 + \cdots + x^n + \mathcal{O}(x^{n+1})$
$\ln(1+x)$	$x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots + \frac{(-1)^{n-1}}{n}x^n + \mathcal{O}(x^{n+1})$
$\arctan(x)$	$x - \frac{1}{3}x^3 + \frac{1}{5}x^5 + \cdots + \frac{(-1)^n}{2n+1}x^{2n+1} + \mathcal{O}(x^{2n+2})$

Goniometrische identiteiten / Trigonometric identities

$$\sin(x+y) = \sin(x)\cos(y) + \cos(x)\sin(y) \qquad \sin^2(x) = \tfrac{1}{2} - \tfrac{1}{2}\cos(2x)$$
$$\cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y) \qquad \cos^2(x) = \tfrac{1}{2} + \tfrac{1}{2}\cos(2x)$$

Vectoren / Vectors

Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ be vectors in $\mathbb{R}^3$.

Inproduct / Inwendig product / Inner product / Dot product:

$$\mathbf{a} \bullet \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$$

Uitproduct / Uitwendig product / Cross product / Vector product:

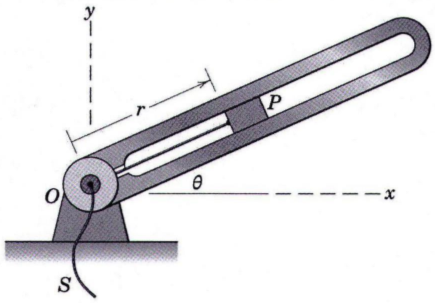
$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}$$

Coordinate transformations:
Cylindrical coordinates: For $r \geq 0$, $\theta \in [0, 2\pi]$: $x = r\cos\theta$, $y = r\sin\theta$, $z = z$
Spherical coordinates: For $R \geq 0$, $\phi \in [0,\pi]$, $\theta \in [0, 2\pi]$:
$x = R\sin\phi\cos\theta$, $y = R\sin\phi\sin\theta$, $z = R\cos\phi$.

Coordinate systems

2p **1**

Consider a slider P in a slotted arm, as depicted in the figure below. The slider can be moved inward by string S, while the slotted arm rotates around point O. The angular position θ of the arm is prescribed as $\theta = 0.5t$ (in rad), where t indicates time. The position of slider P inside the slot can be described as a function of time as $r = 2 - 0.1t$ (in m).



$\dot{\theta} = 0.5$

The motion of slider P is best described in cylindrical coordinates. What is the velocity v of slider P at $t = 2$ s?

$\hat{r}$ $\quad \hat{\theta}\theta$

$\dot{r}\hat{r} + r\dot{\theta}\hat{\theta}$

-0.1

$t=2$

$(2 - 0.1t)$

$(2 - 0.1t).(0.5)$

$1.8 \quad = 0.9$

- (a) $\mathbf{v} = -0.1\hat{\boldsymbol{r}}$
- (b) $\mathbf{v} = 1.8\hat{\boldsymbol{r}} + 0.9\hat{\boldsymbol{\theta}}$
- (c) $\mathbf{v} = 1.8\hat{\boldsymbol{r}} + \hat{\boldsymbol{\theta}}$
- (d) $\mathbf{v} = -0.1\hat{\boldsymbol{r}} + 0.5\hat{\boldsymbol{\theta}}$
- (e) $\mathbf{v} = 1.8\hat{\boldsymbol{r}} + 0.5\hat{\boldsymbol{\theta}}$
- (f) $\mathbf{v} = -0.1\hat{\boldsymbol{r}} + 0.9\hat{\boldsymbol{\theta}}$
- (g) $\mathbf{v} = -0.1\hat{\boldsymbol{r}} + \hat{\boldsymbol{\theta}}$
- (h) $\mathbf{v} = 1.8\hat{\boldsymbol{r}}$

Quadric surface

$x^2 + \frac{(y-1)^2}{2} - (z+4)^2 - 16 = c$

2p **2** Consider a surface in $\mathbb{R}^3$ described by the equation

$$x^2 + \frac{(y-1)^2}{2} - z^2 + 8z = c.$$

with x, y, z real variables, and c a real parameter. Which of the following claims is true?

$x^2 + \frac{(y-1)^2}{2} - (z+4)^2 = c + 16$

- (a) If $c = 10$, then the surface is an hyperboloid of two sheets.
- (b) If $c= 17$, then the surface is a cone.
- (c) If $c = 1$, then the surface is an elliptic paraboloid.
- (d) If $c = 0$, then the surface is an ellipsoid. x
- (e) If $c = 15$, then the surface is an hyperboloid of one sheet.

$z = r$
$\theta = \pi/3$

Coordinate systems

2p **3** Point P has spherical coordinates $[4, \phi, \theta]$ and cylindrical coordinates $[r, \pi/3, r]$. Let $[x, y, z]$ be the Cartesian coordinates of P. Then the x-component of the Cartesian coordinates of P satisfies

- (a) $x = \sqrt{2}$
- (b) $x = \sqrt{3}$
- (c) $x = 1$
- (d) $x = 2$
- (e) $x = \sqrt{6}$
- (f) $x = \sqrt{8}$

$x = r\cos\theta$ $\quad x = \sin\phi\cos\theta$

$4\cos\left(\frac{\pi}{3}\right)$

$4\cos(60^\circ)$

$\frac{4}{2} = 2$

$(1-x)(y-\ln(x)) \geq 0$

Domain of definition

2p **4** Consider the function

$$f(x,y) = \sqrt{(1-x)(y-\ln(x))}.$$

Let $D \subset \mathbb{R}^2$ be the maximal domain of definition. Which of the following claims is true?

- (a) The domain D is closed and $(1,0)$ is an isolated point of the domain.
- (b) The domain D is neither open nor closed and $(1,0)$ is an isolated point of the domain.
- (c) The domain D is neither open nor closed and $(1,0)$ is not an isolated point of the domain.
- (d) The domain D is open and $(1,0)$ is an isolated point of the domain.
- (e) The domain D is open and $(1,0)$ is not an isolated point of the domain.
- (f) The domain D is closed and $(1,0)$ is not an isolated point of the domain.

Arc length

2p **5** What is the length of the curve C, parameterized by

$$\underline{r}(t) = (\sin(t) - t\cos(t))\underline{i} + (\cos(t) + t\sin(t))\underline{j} + t^2\underline{k}, \quad (t \in [-\pi, \pi]).$$

- (a) $\frac{1}{2}\pi^2\sqrt{2}$
- (b) $2\pi\sqrt{5}$
- (c) $\pi\sqrt{2}$
- (d) $\pi^2\sqrt{2}$
- (e) $2\pi\sqrt{2}$
- (f) $\pi\sqrt{5}$
- (g) $\frac{1}{2}\pi^2\sqrt{5}$
- (h) $\pi^2\sqrt{5}$

Tangent line

2p **6** The intersection of the plane $x + y = 1$ and the surface $z = x^2y + y^2x$ is a curve. What is the correct vector representation of the tangent line to this curve in the point $(2, -1, -2)$?

a) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 1 \\ 3 \end{bmatrix}, \quad (\lambda \in \mathbb{R}).$

b) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad (\lambda \in \mathbb{R}).$

c) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad (\lambda \in \mathbb{R})$

d) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ 7 \end{bmatrix}, \quad (\lambda \in \mathbb{R})$

e) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}, \quad (\lambda \in \mathbb{R})$

f) $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad (\lambda \in \mathbb{R})$

Chain rule

2p **7** Suppose that $z = f(x,y)$, and $x = 2s + 4t$ and $y = 3s - 2t$. Then $\frac{\partial^2 z}{\partial t^2}$ is equal to

- (a) $4f_{11} + 12f_{12} + 9f_{22}$
- (b) $16f_{11} - 16f_{12} + 4f_{22}$
- (c) $16f_{11} + 8f_{12} + 6f_{22}$
- (d) $9f_{11} - 16f_{12} + 4f_{22}$
- (e) $9f_{11} - 12f_{12} + 4f_{22}$
- (f) $4f_{11} - 6f_{12} + 2f_{22}$

Limit

2p **8** Consider the function $f(x,y) = \frac{x^2(y-1)^2}{x^2 + (y-1)^2}$. The limit

$$\lim_{(x,y)\to(0,1)} f(x,y)$$

- (a) does not exist
- (b) depends on the direction of approach
- (c) is $+\infty$
- (d) is $-\infty$
- (e) is 0
- (f) is 1

$\underline{x=0}$ $\quad$ $\underline{y=1}$

$\frac{0}{(y-1)^2}$ $\quad$ $\frac{0}{x^2}$

Directional derivative

2p **9** What is the directional derivative of the function $f(x,y) = \dfrac{x}{1+y}$

in the point $(2,0)$ in the direction of the vector $\underline{u} = \underline{i} - \underline{j} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$?

ⓐ $D_{\underline{u}}f(2,0) = \frac{1}{2}$

$= \nabla f \vec{v}$

ⓑ $D_{\underline{u}}f(2,0) = -\frac{1}{2}\sqrt{3}.$

ⓒ $D_{\underline{u}}f(2,0) = \sqrt{3}$

ⓓ $D_{\underline{u}}f(2,0) = \frac{3}{2}\sqrt{2}$

ⓔ $D_{\underline{u}}f(2,0) = -\frac{1}{2}\sqrt{2}$

ⓕ $D_{\underline{u}}f(2,0) = 2$

ⓖ $D_{\underline{u}}f(2,0) = \sqrt{2}$

ⓗ $D_{\underline{u}}f(2,0) = -3$

Tangent plane

2p **10** Let S be the level surface of the function

$$f(x,y,z) = 2yz\cos(x) - e^{y-1}z$$

that passes through the point $(0,1,2)$, i.e. S is a surface in $\mathbb{R}^3$. The tangent plane to S in the point $(0,1,2)$ is given by the equation

$\nabla f =$

ⓐ $z = -2y$

$4(y-1) + (z-2) = 2$

ⓑ $w = 2(y-1) + (z-2)$

$4y - 4 + z - 2 - 2 = 0$

ⓒ $w = -4x - 2(y-1) - (z-2)$ ×

ⓓ $4x + 2y + z = 4$ ×

ⓔ $x - 2y - z = -4$ ×

$4y - \cancel{4} + z - 2 = 0$

ⓕ $4(y-1) + 2(z-2) = 0$

$4y + z - 6 = 0$